MobileComp: Real-Time Full Projector Compensation on Mobile Devices

— Supplementary Materials —

Category: Research

1 Supplementary Method Details

These supplementary details expand the formulations summarized in the main paper. We first define the full-compensation objective and projector-domain warp, and then specify the reusable surface cache and the training objectives used to adapt MobileComp and SI-MobileComp.

1.1 Full projector compensation

Following the full projector compensation formulation in the main paper, we decompose the project-and-capture process of setup s into photometric transformation $\mathcal{H}_s$ and geometric transformation $\mathcal{G}_s$:

$$\widetilde{x}_{s,i} = \mathcal{G}_s(\mathcal{H}_s(x_i)). \tag{S1}$$

Full projector compensation seeks projector input $x^*_{s,T}$ whose camera-captured projection reproduces desired frame x'_T:

$$\mathcal{G}_s\big(\mathcal{H}_s(x^*_{s,T})\big) = x'_T. \tag{S2}$$

Because the coupled transformation has no closed-form inverse, the main paper uses its pseudo-inverse decomposition to separate geometric correction from photometric compensation.

1.2 Projector-domain preprocessing and native warp

Following the preprocessing described in the main paper, we subtract a dark reference from the projection surface captured under a uniform bright input, convert the difference to luminance, and apply Gaussian smoothing and Otsu thresholding. The convex hull of the largest contour defines the FOV mask M_s. Its bounding-box corners initialize GANet's affine parameters θ_s^{aff}, and the largest inscribed 1:1 region R_s defines the clipping-free output area. If $(t_{x,s}, t_{y,s})$ and ℓ_s denote the top-left corner and side length of R_s, the homogeneous transform A_s maps the canonical grid into this region, while A_s^{-1} normalizes camera samples to the canonical domain. GANet's refinement stages or the FlowFormer estimator described in the main paper then estimate the residual non-affine mapping.

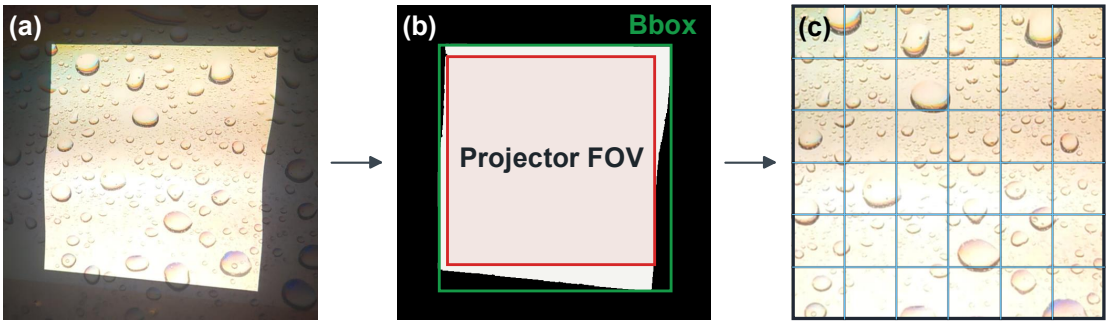


Figure S1: Projector FOV identification and affine normalization on the canonical square output canvas. **(a)** Camera capture of the projection surface under a uniform bright projector input. **(b)** Binary projector FOV mask (white), with the axis-aligned bounding box in green and the largest clipping-free 1:1 displayable area in red. **(c)** Surface samples within the red area normalized to the canonical domain; the blue grid denotes the regular sampling lattice after applying A_s^{-1}.

Let Γ_s denote the sampling grid of either deployment protocol. For output pixel p, the native bilinear warp is

$$\begin{aligned} \mathcal{W}(I;\Gamma_s)(p) &= \sum_{q\in\mathcal{N}(\Gamma_s(p))} w_{s,p,q} I(q), \\ \sum_{q\in\mathcal{N}(\Gamma_s(p))} w_{s,p,q} &= 1, \end{aligned} \tag{S3}$$

where $\mathcal{N}(\Gamma_s(p))$ contains the four neighboring source pixels. We precompute their indices and weights once per setup.

1.3 Surface reference cache and memory

SI-MobileComp captures five intensity-varying surface references at projector intensities $K = \{1, 32, 63, 94, 125\}$. For a cache with L feature scales, its tensor footprint is

$$M_{\mathcal{F}} = b \sum_{\ell=1}^{L} C_\ell H_\ell W_\ell, \tag{S4}$$

where C_ℓ, H_ℓ, and W_ℓ define the ℓth feature shape and b is the number of bytes per feature value. This expression accounts only for persistent cached features; model weights, intermediate tensors, media buffers, and runtime workspace are included separately in the measured peak RAM.

1.4 MobileComp training objective

Let $\widehat{x}_{S,s,i}$ and $\widehat{x}_{T,s,i}$ denote the MobileComp student and frozen FP32 CompenHR teacher predictions for training pair i in setup s. The component losses in the main-paper objective are

$$\begin{aligned}\mathcal{L}_{\text{task}} &= \|\widehat{x}_{S,s,i} - x_i\|_1 + \|\widehat{x}_{S,s,i} - x_i\|_2^2 \\ &\quad + 1 - \text{SSIM}(\widehat{x}_{S,s,i}, x_i), \\ \mathcal{L}_{\text{out}} &= \|\widehat{x}_{S,s,i} - \widehat{x}_{T,s,i}\|_1, \\ \mathcal{L}_{\text{color}} &= \Delta E_{00}\big(\text{Lab}(\widehat{x}_{S,s,i}), \text{Lab}(x_i)\big).\end{aligned} \tag{S5}$$

All pixel losses use mean reduction, and ΔE_{00} is averaged over the image.

Let Z_S^ℓ and Z_T^ℓ denote the student and teacher activations, respectively, at the ℓth selected PANet stage. To compare activations with unequal channel widths, we convert $Z \in \mathbb{R}^{C\times H\times W}$ into the normalized channel-energy map

$$\mathcal{E}(Z) = \frac{C^{-1}\sum_{c=1}^{C} Z_c^2}{\text{mean}_p\big(C^{-1}\sum_{c=1}^{C} Z_c^2(p)\big) + 10^{-6}}, \tag{S6}$$

and define

$$\begin{aligned}\mathcal{L}_{\text{feat}} &= \frac{1}{3}\sum_{\ell\in\mathcal{Q}} \left\| \mathcal{E}(Z_S^\ell) - \mathcal{E}(Z_T^\ell) \right\|_1, \\ \mathcal{Q} &= \{\texttt{conv2}, \texttt{conv3}, \texttt{conv5}\}.\end{aligned} \tag{S7}$$

For training iteration $k \in \{0, \ldots, K_{\text{MC}} - 1\}$, the scheduled weights are

$$\begin{aligned}&u_k = \frac{k}{K_{\text{MC}} - 1}, \qquad \alpha_k = 0.5 + 0.5u_k, \qquad \beta_k = 1.6 - 1.1u_k, \\ &w_{\text{task}} = \alpha_k, \quad w_{\text{out}} = \beta_k + 0.30, \quad w_{\text{feat}} = 0.08, \quad w_{\text{color}} = 0.03.\end{aligned} \tag{S8}$$

Thus, the effective output-distillation coefficient decreases from 1.90 to 0.80 as the supervised task weight increases.

1.5 SI-MobileComp training objective

For paired projector input x_i and camera capture $\widetilde{x}_{s,i}$, the surrogate inverse formulation in the main paper defines

$$\widetilde{x}_{s,i} = \mathcal{G}_s(\mathcal{H}_s(x_i)), \qquad \widehat{x}_{s,i} = R_\theta\Big(\mathcal{W}(\widetilde{x}_{s,i}; \Omega_s), F_s^1, F_s^2, F_s^3\Big). \tag{S9}$$

Let $\mathcal{B}$ be a mixed batch of paired and teacher-only samples and $\mathcal{B}_{\text{p}} \subseteq \mathcal{B}$ its paired subset. For sample j, $\widehat{x}_{S,j}^*$ and $\widehat{x}_{T,j}^*$ are the student and teacher predictions, and x_j is available only for $j \in \mathcal{B}_{\text{p}}$. We define

$$\begin{aligned}\ell_{\text{sup},j} &= \|\widehat{x}_{S,j}^* - x_j\|_1 \\ &\quad + 0.2\big(1 - \text{SSIM}(\widehat{x}_{S,j}^*, x_j)\big), \\ \mathcal{L}_{\text{sup}}(\mathcal{B}_{\text{p}}) &= \begin{cases} \dfrac{1}{|\mathcal{B}_{\text{p}}|}\displaystyle\sum_{j\in\mathcal{B}_{\text{p}}} \ell_{\text{sup},j}, & |\mathcal{B}_{\text{p}}| > 0, \\ 0, & |\mathcal{B}_{\text{p}}| = 0. \end{cases}\end{aligned} \tag{S10}$$

which removes the undefined normalization for an empty paired subset. The distillation terms are

$$\begin{aligned}\mathcal{L}_{\text{out}}(\mathcal{B}) &= \frac{1}{|\mathcal{B}|}\sum_{j\in\mathcal{B}} \|\widehat{x}_{S,j}^* - \widehat{x}_{T,j}^*\|_1, \\ \mathcal{L}_{\text{surf}}(\mathcal{B}) &= \frac{1}{3|\mathcal{B}|}\sum_{j\in\mathcal{B}}\sum_{\ell=1}^{3} \|H_\ell(F_{S,j}^\ell) - F_{T,j}^\ell\|_1.\end{aligned} \tag{S11}$$

Substitution into the compact main-paper objective gives the original expanded loss. H_1 is identity; the training-only 1×1 heads H_2 and H_3 map 24/48 student channels to 64/128 teacher channels and are discarded after training.

2 MOBILE IMPLEMENTATION DETAILS

The mobile application validates the reusable setup state before entering the per-frame path and exposes the corresponding initialization and runtime controls.

2.1 Setup-package validation

During SI-MobileComp initialization, the application writes Ω_s and the three tensors in $\mathcal{F}_s$ to the setup package. It verifies the exact byte size of every tensor before writing a completion marker. The real-time profile accepts the package only after this marker is present, preventing stale or partially serialized state from entering the per-frame path.

2.2 Android application

The interface integrates setup initialization, image and video inputs, asynchronous playback, and runtime logging in the same smartphone workflow.

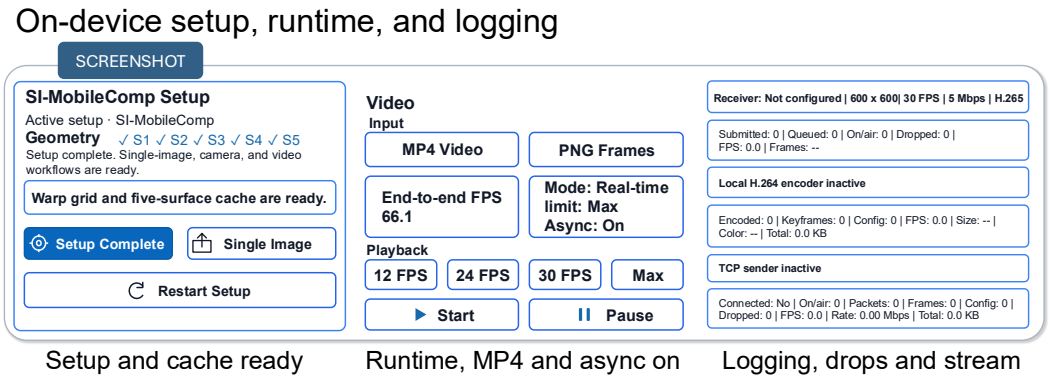


Figure S2: Android interface for smartphone-side setup initialization, runtime control, and logging. The interface reports geometry and surface-cache readiness, selects image or video inputs and asynchronous playback, and monitors encoding, streaming, throughput, and dropped frames.

3 Additional Qualitative Results

To complement the still-image comparisons in the main paper, we provide successive compensation frames from two Blender Open Movies. *Spring* contains richly textured natural scenes, whereas *Wing It!* contains saturated animated content with large inter-frame appearance changes. Together, Figs. S3 and S4 show how the same setup package is reused as the desired video content changes over time.



Figure S3: Video compensation on successive frames from the Blender Open Movie *Spring* (Blender Foundation, CC BY 4.0). Columns progress in time; rows show uncompensated camera captures, desired frames, and compensated camera captures. The inset shows the projection surface.

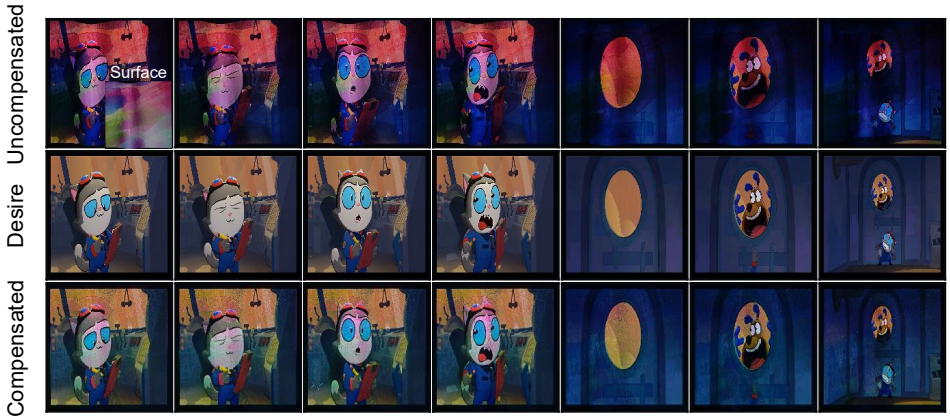


Figure S4: Video compensation on successive frames from the Blender Open Movie *Wing It!* (Blender Foundation, CC BY 4.0). Columns progress in time; rows show uncompensated camera captures, desired frames, and compensated camera captures. The inset shows the projection surface.